IGCSE · Cambridge (CIE) · Maths

45 mins   13 questions

Non-Calculator Questions

Quadratic Equations

Solving Quadratics by Factorising / The Quadratic Formula / Solving Quadratics using a Calculator

Medium (7 questions)	/21
Hard (3 questions)	/9
Very Hard (3 questions)	/15
Total Marks	**/45**

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Medium Questions

1 Solve $x^2 - 2x - 6 = 0$.

Give your answer in the form $a \pm \sqrt{b}$ where a and b are integers.

(3 marks)

2 Factorise.

$1 + x - y - xy$

(2 marks)

3 (a) Write $6x^2 - 7x - 3$ in the form $(2x + a)(3x + b)$, where a and b are integers.

(2 marks)

(b) Hence, or otherwise, solve $6x^2 - 7x - 3 = 0$.

(2 marks)

4 Solve.

$6x^2 - 5x - 6 = 0$

$x =$ .... or $x =$ ....

(3 marks)

5 Solve.

$$8x^2 - 2x - 3 = 0$$

$x =$ .... or $x =$ ....

(3 marks)

6 Solve $2x^2 + 5x - 12 = 0$.

(3 marks)

7 Solve.

$$x^2 - 6x + 1 = 0$$

Give your answers in the form $p \pm q\sqrt{2}$ where p and q are constants to be found.

(3 marks)

Hard Questions

1 The solutions to the equation $x^2 + gx + h = 0$ are $\frac{1 - \sqrt{17}}{2}$ and $\frac{1 + \sqrt{17}}{2}$.

Find the value of g and the value of h.

(3 marks)

2 Solve by factorisation $10r^2 - 23r + 9 = 0$.

r = ... or r = ...

(3 marks)

3 $x^2 - 12x + a = (x + b)^2$

Find the value of a and the value of b.

a = ...
b = ...

(3 marks)

Very Hard Questions

1 The solutions of the equation $x^2 + bx + c = 0$ are $\frac{-7+\sqrt{61}}{2}$ and $\frac{-7-\sqrt{61}}{2}$.

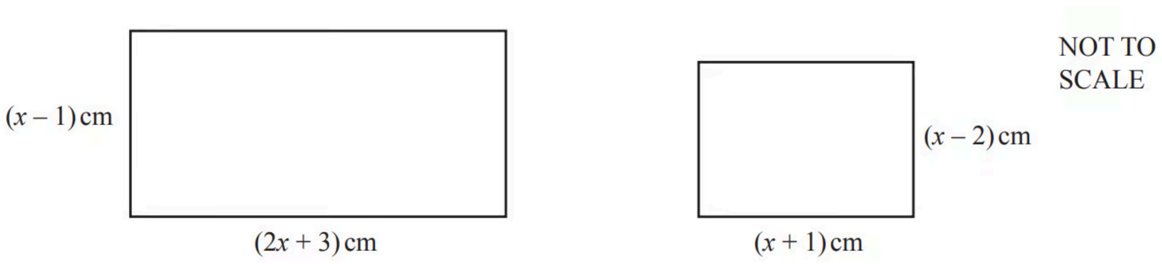
Find the value of b and the value of c.

$b =$ ....

$c =$ ....

(3 marks)

2



The difference between the areas of the two rectangles is 62 cm^2.

i) Show that $x^2 + 2x - 63 = 0$.

[3]

ii) Factorise $x^2 + 2x - 63$.

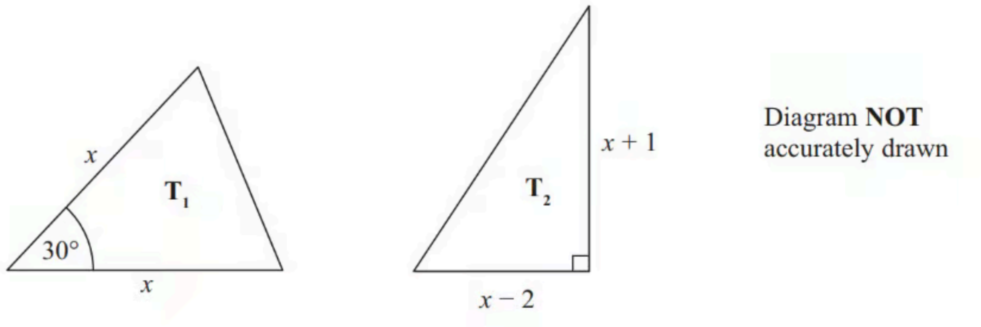
[2]

iii) Solve the equation $x^2 + 2x - 63 = 0$ to find the difference between the perimeters of the two rectangles.

.... cm [2]

(7 marks)

3



The lengths of the sides are in centimetres.
The area of triangle $\mathbf{T}_1$ is equal to the area of triangle $\mathbf{T}_2$.

Work out the value of x, giving your answer in the form $a+\sqrt{b}$ where a and b are integers.

(5 marks)